Completed impact run: targets not reproduced

The declared 20 ms calculation completed. It did not reproduce Bowen’s beam response or the experimental deflection. The computed fixed bottom probe reaches 253.61 mm and is still moving downward; the raw impact-force peak is not converged.

Report generated 2026-10-11 12:39:33 UTC+08. The production calculation completed 66,667 steps through 20.0001 ms. All final numerical states and all 666 scheduled energy samples are finite. Completion of this interval does not establish a physical displacement maximum.

Case / evidence	Peak force [kN]	Probe [mm]	Window [ms]
Bowen, Table 3-1	359.800	22.700	Published maxima
Experiment cited by Bowen	301.000	23.380	Published maxima
Reconstruction, dt 0.30 µs	272.977	253.613	20.0001
Reconstruction, dt 0.15 µs	278.938	33.218	3.000
Reconstruction, dt 0.075 µs	293.642	2.213	.300
Reconstruction, dt 0.0375 µs	362.702	2.213	.300
Recovered original model	220.663	38.089	3.000

Computed probe values are finite-window motion of fixed concrete particles; their equivalence to the experimental measurement is unknown. None is a validated maximum deflection. At 20 ms no bottom probe and no steel node remains connected to a vertical bearing in the mechanical load-transfer graph.

The production raw force is -9.31% from 301.0 kN and -24.13% from 359.8 kN. The finest-step 362.70 kN peak is a verified discrete-contact activation spike; its proximity to Bowen is not agreement.

No material parameter, artificial damping, contact gap or force filter was fitted to reported peaks. The force, fragment motion and crack morphology must be assessed together. The report documents what was corrected, what is source-supported, and what remains unidentified.

Published inputs and equation-to-code audit

Quantity	Reconstructed input
Beam / support span	600 × 70 × 100 mm / 500 mm
Particle spacing / horizon	5 mm / 15 mm; 33,600 concrete particles
Concrete	E 25 GPa; ft 4.6 MPa; fc 60 MPa; ρ 2400 kg/m³
Fracture energy / steel	105 J/m²; E 200 GPa; fy 482 MPa; ρ 7750 kg/m³
Longitudinal bars	Two 6 mm bars; centre 20 mm above beam bottom
Impactor	38.4 kg; 10.8 m/s; hemispherical tip radius 60 mm

Cell-centred concrete coordinates are combined with 270 steel nodes and 268 axial rods. Full concrete volume plus embedded steel mass is retained; concrete displaced by the bars is not subtracted in this declared hybrid representation.

Concrete force and Eq. 2-12 kernel

Each unique pair applies equal opposite physical forces c(r) × bond-law stretch × Vi Vj × volume/surface factors [N]. Micromodulus has units N/m⁶; acceleration divides by physical nodal mass.

PDROD steel–concrete transfer

122 virtual spherical samples per steel node use eight-corner interpolation. Weighted opposing concrete forces preserve net force and torque, and match interpolation virtual work. This proves conservation, not the missing interface strength/stiffness inputs.

Steel / Eqs. 2-45–47 contact

Corotational axial rods use elastic-perfectly-plastic return mapping. Sphere relocation changes nodal momentum; reaction is impulse/dt. The hammer receives equal recoil. Its horizontal guide impulse is recorded separately.

Damage / measurement

Eq. 2-10 cumulative failed-family volume is distinct from current force inactivity and crack opening. Connectivity excludes analytically zero horizon-shell interactions. Concrete, steel and hammer probes are reported separately.

References: thesis Chapters 2–3, PDROD Section 2.5 and Appendices A–C. Printed p. 32 / PDF p. 49 gives the target table. Contact is printed pp. 27–28 / PDF pp. 44–45. Fig. 3-4 is force versus deflection, not a recovered experimental time history.

Corrections, scientific alternatives and unknowns

Physical units and PDROD action/reaction were checked directly. Contact reaction contains the physical mass and the new-time velocity change, exactly as printed in Eqs. 2-45–47. No remaining concrete-contact extra-volume, extra-dt, sign or velocity-reference error was found.

Fracture work: the printed terminal-stretch formula belongs to a constant kernel, while the selected micromodulus is weighted. Their combination is scientifically inconsistent. It is not a proven transcription error in the unavailable author executable.

$\Gamma = (\pi/2)\, s_0\, s_c \int_0^{\delta} c(r)\, r^4\, dr$
Constant kernel: $\Gamma = (6/5)\, E\, \delta\, s_0\, s_c$
Weighted kernel: $\Gamma = (32/35)\, E\, \delta\, s_0\, s_c$

Declared terminal definition	sc	Lattice mean Γ
Printed constant + weighted kernel	.001268115942	77.8259 J/m²
Weighted continuum work	.001664402174	102.1465 J/m²
Current weighted lattice mean	.001710898586	105.0000 J/m²

The continuum weighted/constant work ratio is 16/21. The terminal correction follows that integral and the independently calculated source lattice calibration; it does not use impact peaks. An orientation mean of 105 J/m² does not make all cleavage planes equal to 105.

Rate alternative: the printed join at 1/s drops the strain-threshold multiplier by 28.99%. Production uses a continuous join at 1/s with the printed branches. The printed coefficient is alternatively continuous at 3/s; Appendix A has another exponent. The original author’s chosen policy is unknown.

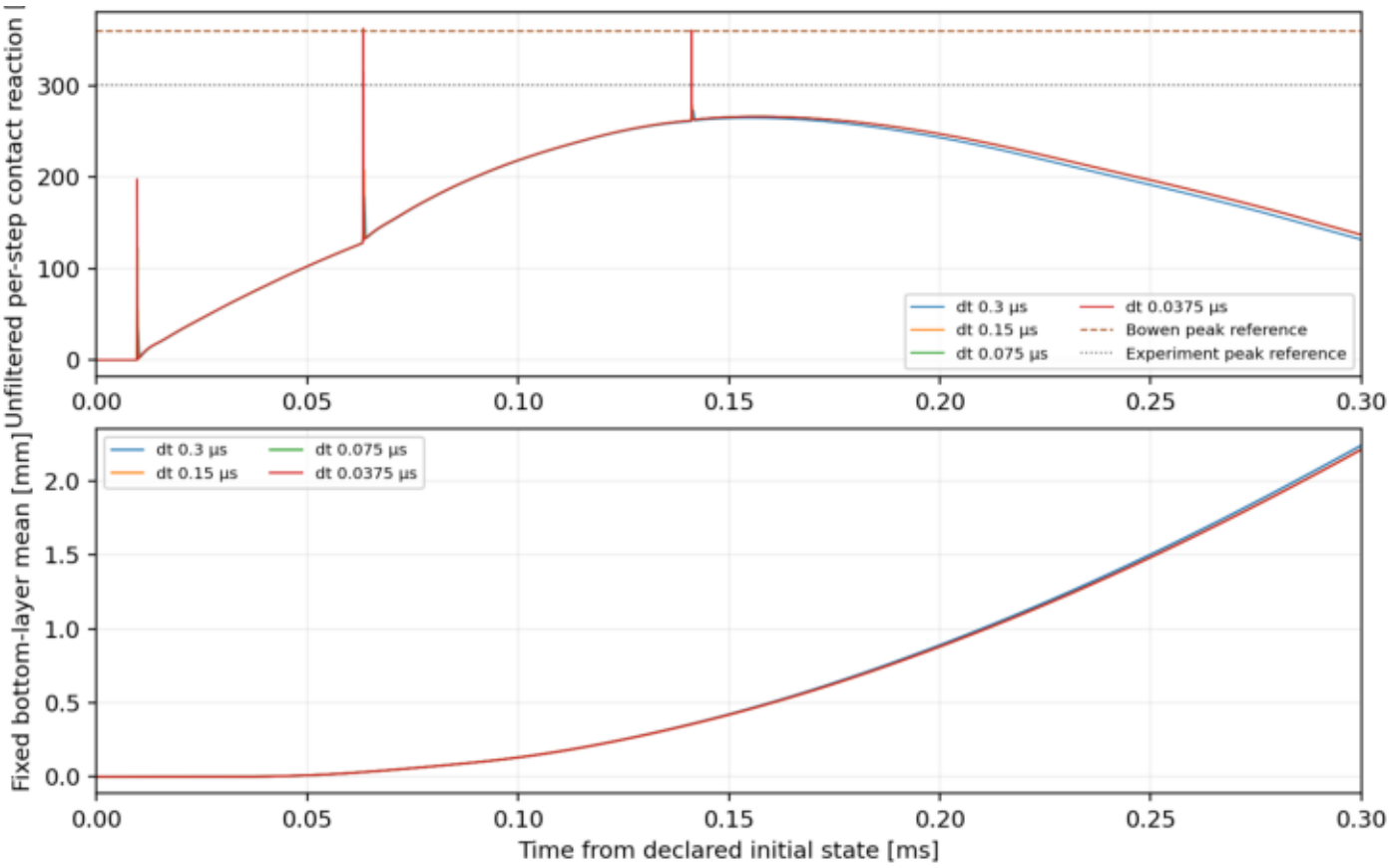
Other declared choices: irreversible history; static micromodulus; particle-top-tangent initial contact; concrete-weighted static interface law. No dynamic micromodulus, tuned smoothing or tuned contact stiffness is enabled.

Unrecovered definitions include interface bcc/s01/sc1/control volumes, bar ends and lateral positions, 60 mm hook rise, 10 mm bilateral bearing patches and anchors, original integration loop, hammer body, contact origin, rate history and sensor processing. The supplied different-case Fortran cannot identify these Chapter 3 inputs.

Isolated 3 ms interface-history sensitivity: ignoring previous failure revives 4,379 samples, including 1,174 in tension. The sum of nodal steel-force magnitudes changes from 8.09 kN to 3.413 MN; force-primitive potential changes from .015 to 1920 J. Both branches conserve force and torque. These are evaluations at one geometry, not a new trajectory or accepted correction. Appendix C omits gating but the full author loop is missing; tensile restoration permits healing and mid-run policy switching would insert energy. Failed PDROD samples currently lose all forces, with no separate steel–concrete closure contact.

Timestep refinement: raw peak does not converge

dt [µs]	Raw peak [kN]	Peak time [ms]	Impulse ≤.3 ms [N·s]
0.3	272.977271	0.1416000	54.731091
0.15	278.937795	0.1413000	55.420588
0.075	293.642356	0.1411500	55.438647
0.0375	362.701700	0.0632625	55.442454



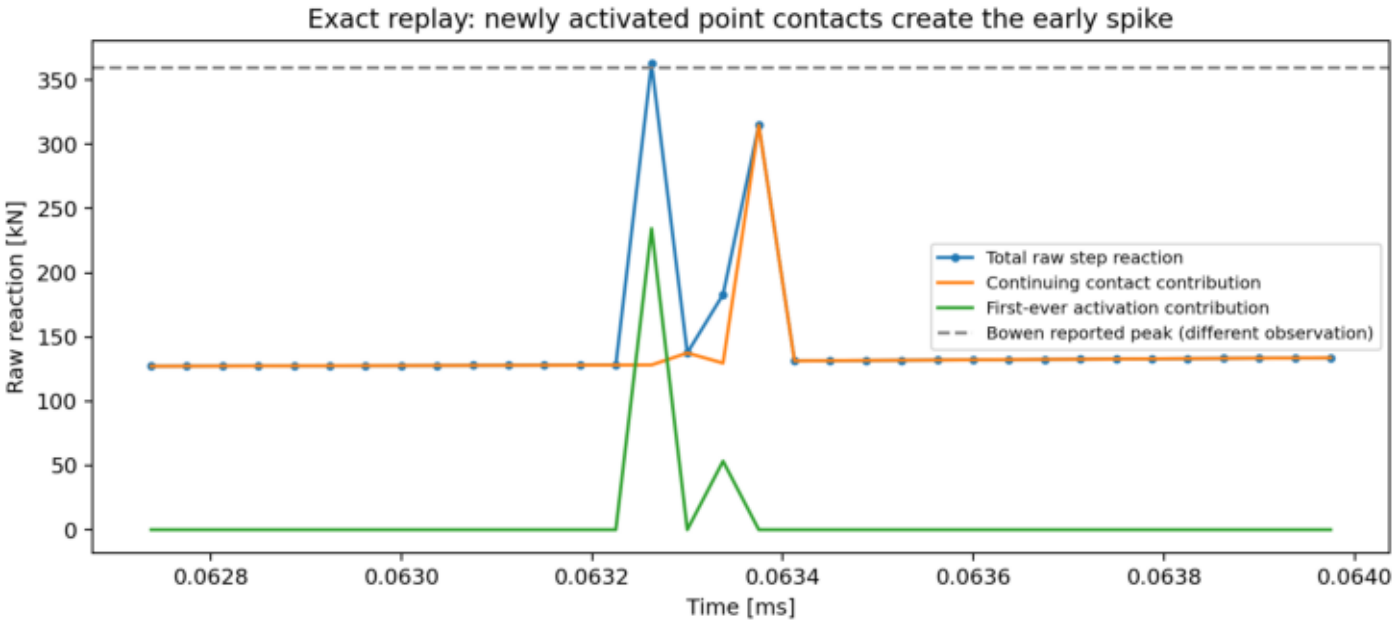
The raw peaks change by +2.18%, +5.27% and +23.52% under successive dt halving. At the finest dt the controlling peak moves from approximately 0.141 ms to 0.0632625 ms. These values do not support a finite extrapolated force maximum.

Coarse and half-dt share identical frozen v3 inputs except dt. At 3 ms, bottom motion is 33.906416 versus 33.217723 mm (-2.031%). Impulse is 65.085371 versus 66.431531 N·s.

Temporal sensitivity is separate from full impact mesh convergence, which has not been established. The nearly identical early displacement at the smallest steps does not repair the unfiltered force observation or the severe later structural response.

Verified contact activation spike: exact replay

All numeric fields of all 1,750 replayed rows match the retained original run exactly. The unchanged operator at step 1687, 0.0632625 ms gives 362.701700 kN. Four newly contacting nodes contribute 234.443369 kN; four continuing nodes contribute 128.258331 kN.



Each new point has mass 0.0003 kg. Projection changes its predicted vertical velocity from about −2.952887 to −10.279242 m/s. The four finite activation impulses total 0.008791626 N·s; dividing by dt = 3.75×10^{-8} s produces the large per-step sample.

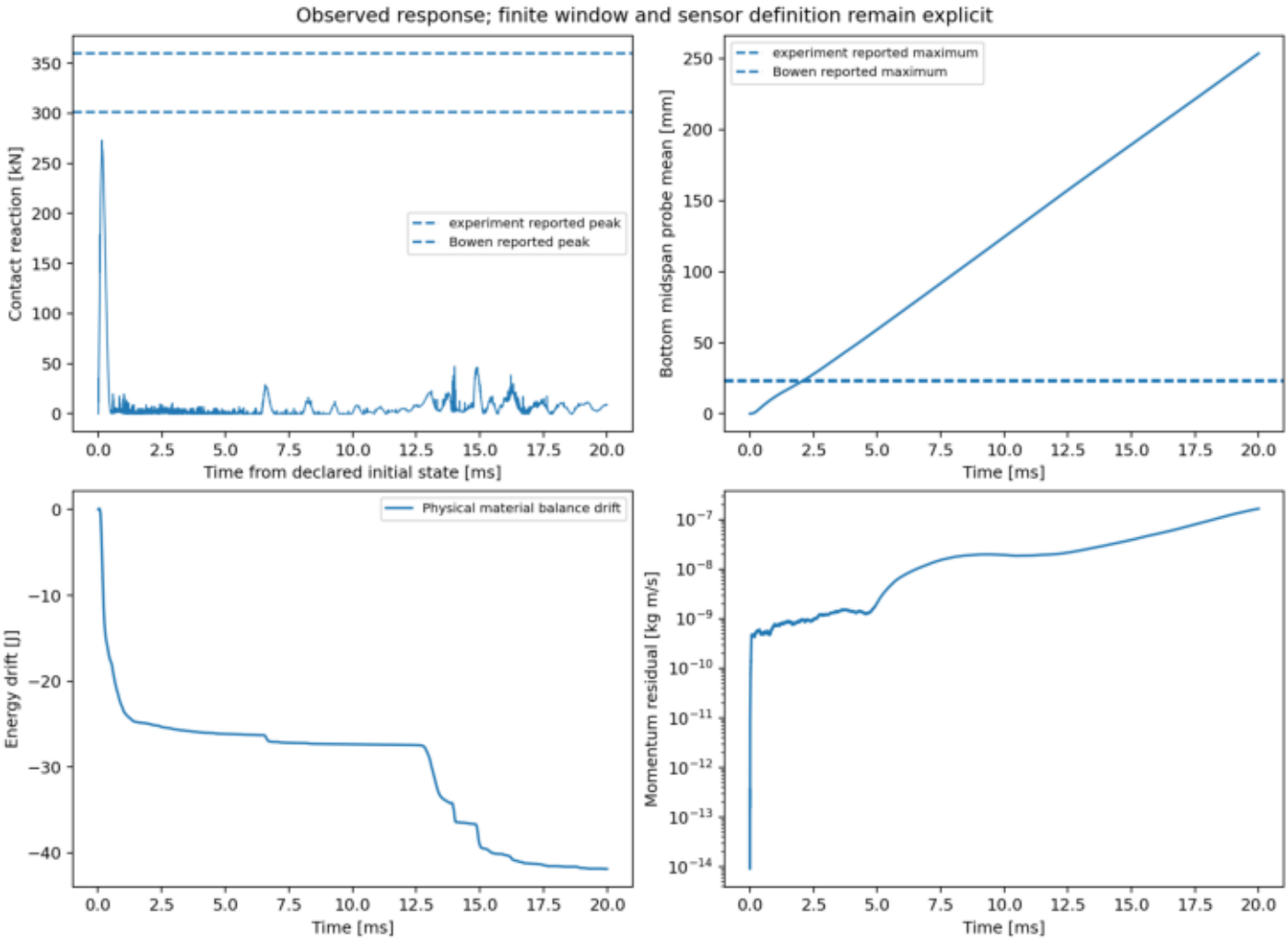
Actual-operator tangent-node verification: Δv = −V; I = mV; Fsample = mV/dt. Four resting points have total mass 0.0012 kg and incoming speed 10.8 m/s.

dt [μs]	Impulse [N·s]	Raw reaction [kN]
0.3	0.01296	43.2
0.15	0.01296	86.4
0.075	0.01296	172.8
0.0375	0.01296	345.6
0.01875	0.01296	691.2

An entry event may split its impulse over two bins. Time-grid phase, normals and pre-contact forces change individual peaks; they do not remove the finite nodal velocity jump. This is a rigid lumped point-contact limitation, not incorrect force normalization.

The experiment’s force bandwidth, sampling and peak-processing rule remain unknown. No supported filter or compliant-contact replacement was found. Refining until a spike approaches 359.8 kN is not reproduction, and no filter or contact parameter was selected to obtain agreement.

Completed 20 ms production response

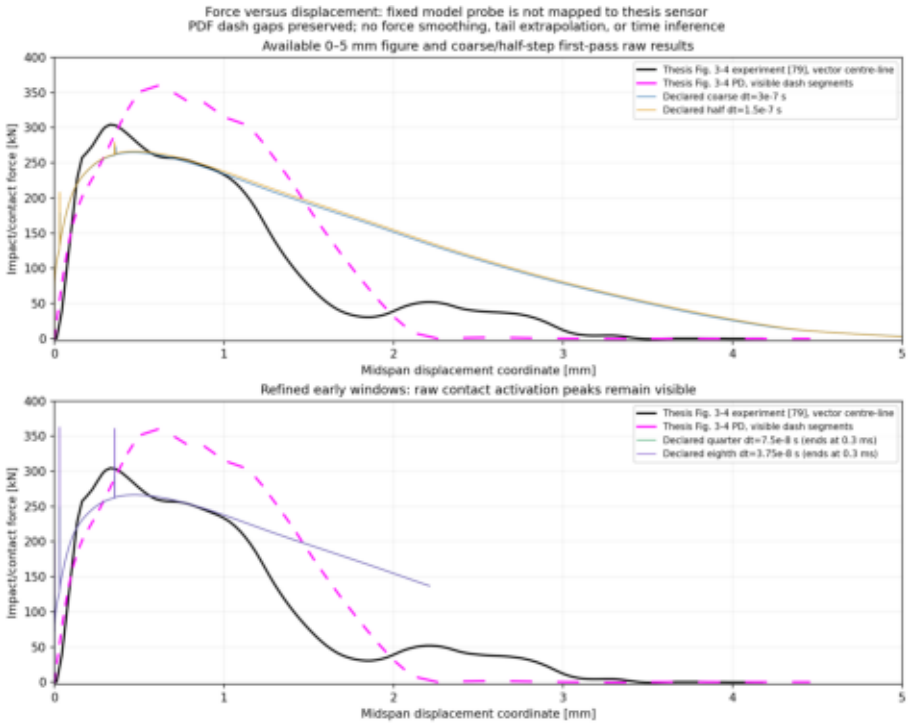


Unfiltered maximum: 272.977271 kN at 0.1416 ms. First positive contact: 0.0099 ms. Integrated impulse through 20.0001 ms: 147.000887 N·s.

Fixed bottom-layer mean at the endpoint: 253.612860 mm, still moving downward at 12.963251 m/s. Final hammer displacement: 169.893807 mm; hammer velocity: −6.971852 m/s. Contact reaction at the endpoint remains approximately 9.217 kN.

The history passed 22.70 and 23.38 mm without reaching a turning point. Stopping at that passage would be an invalid maximum. The 20 ms endpoint is a declared observation window informed by the supplied separate Wang model; it is not Bowen’s recovered stop criterion.

Bowen Fig. 3-4: force–deflection comparison



Source: thesis p. 32 / PDF p. 49. PDF centre-lines preserve 157 black experimental segments and 37 visible PD dashes. Dash gaps remain open; no smoothing, tail extrapolation or force–time history is invented.

Drawn/recorded curve	Peak [kN]	Deflection at peak [mm]
Figure experiment centre-line	304.525	.329650
Figure Bowen PD visible dashes	359.727	.610228
Current dt .30 µs	272.977	.360764
Current dt .0375 µs activation spike	362.702	.030876

Current probe: −mean(u_z) over 28 fixed full-width bottom particles, at x = 297.5/302.5 mm and z = 2.5 mm. Equivalence to Bowen's extraction or the experimental optical ROI is unknown. Reactions are unfiltered; no alignment or processing parameter was fitted.

The 0–5 mm source plot does not establish the complete 22–23 mm event. Coarse/fine traces end at their first 5 mm crossing; finer runs end at .3 ms / 2.213 mm. The eighth-step near-target spike is at only .0309 mm because additional points activate contact.

Drawn experiment apex: 304.525 kN versus Table 3-1's 301.0 kN. Figure legend cites [79], table/text [78]. Extraction estimates the drawn line, not original physical data or uncertainty. Source identity and sensor limitations prevent a matched-observation validation claim.

Fixed displacement probes and structural detachment

Final checkpoint observable	Coarse 20.0001 ms	Half-dt 3 ms
Bottom mean downward [mm]	253.613	33.218
Bottom downward speed [m/s]	12.963	12.181
Support-connected bottom probes	0/28	28/28
Support-connected concrete	56/33,600	31,748/33,600
Support-connected steel	0/270	270/270
Two steel midspan probes [mm]	119.195, 122.548	34.267, 33.612
Detached concrete nodes	33,544	1,852
Maximum any concrete norm [mm]	2125.642	279.751

The graph uses positive-stiffness concrete bonds, positive-weight PDROD links and steel rods, excluding analytically zero horizon-shell interactions. The bearing constraints remain attached to original material IDs: their vertical motion is fixed but final lateral motion reaches 317.019 mm. This permanent material patch differs from spatial contact with a finite stationary bearing. The author’s bearing/friction/anchor definitions are unavailable.

At the production endpoint 33,544 concrete points and all 270 steel nodes are detached from the bearing graph. Maximum any-particle motion exceeds 2.12 m. This is severe collapse/fragment motion, not a bounded 22.70 mm RC beam deflection. A small momentum residual does not validate this structural outcome.

Experimental DIC may follow a separated shear plug, so detachment alone does not disqualify a sensor. The original measured point/ROI and time are still required. Fixed concrete and steel probes remain explicit; a changing supported subset or a maximum displacement norm is never substituted for the reported midspan sensor.

Direct steel contact is omitted here but included in the supplied Fortran. Verified snapshots at 1.5, 3, 6, 12 and 20.0001 ms contain no steel point penetration or bar-radius overlap with the sphere; the final minimum bar-surface clearance is 24.65 mm. No upper-sphere concrete contact appears in those states. This excludes those mechanisms at the inspected snapshots, not at every intervening step.

Energy and momentum: bookkeeping and limits

Physical material balance: D = KE + stored + deleted rupture + steel plastic − initial KE − rate-potential exchange.

Contact is internal to the hammer/material system. Rate updates exchange potential. Contact/support projection effects remain in the physical balance. Subtracting cumulative projection kinetic changes gives a different operator-splitting ledger; it cannot be used to conceal physical drift.

Sampled diagnostic	Coarse ≤20 ms	Half-dt ≤3 ms	Original ≤3 ms
Initial hammer KE [J]	2239.488	2239.488	2239.488
Max |physical drift| [J]	41.856	5.840	2.809
Max drift / initial KE [%]	1.869	0.261	0.125
Max momentum residual [kg m/s]	1.659e-07	4.633e-09	9.529e-10

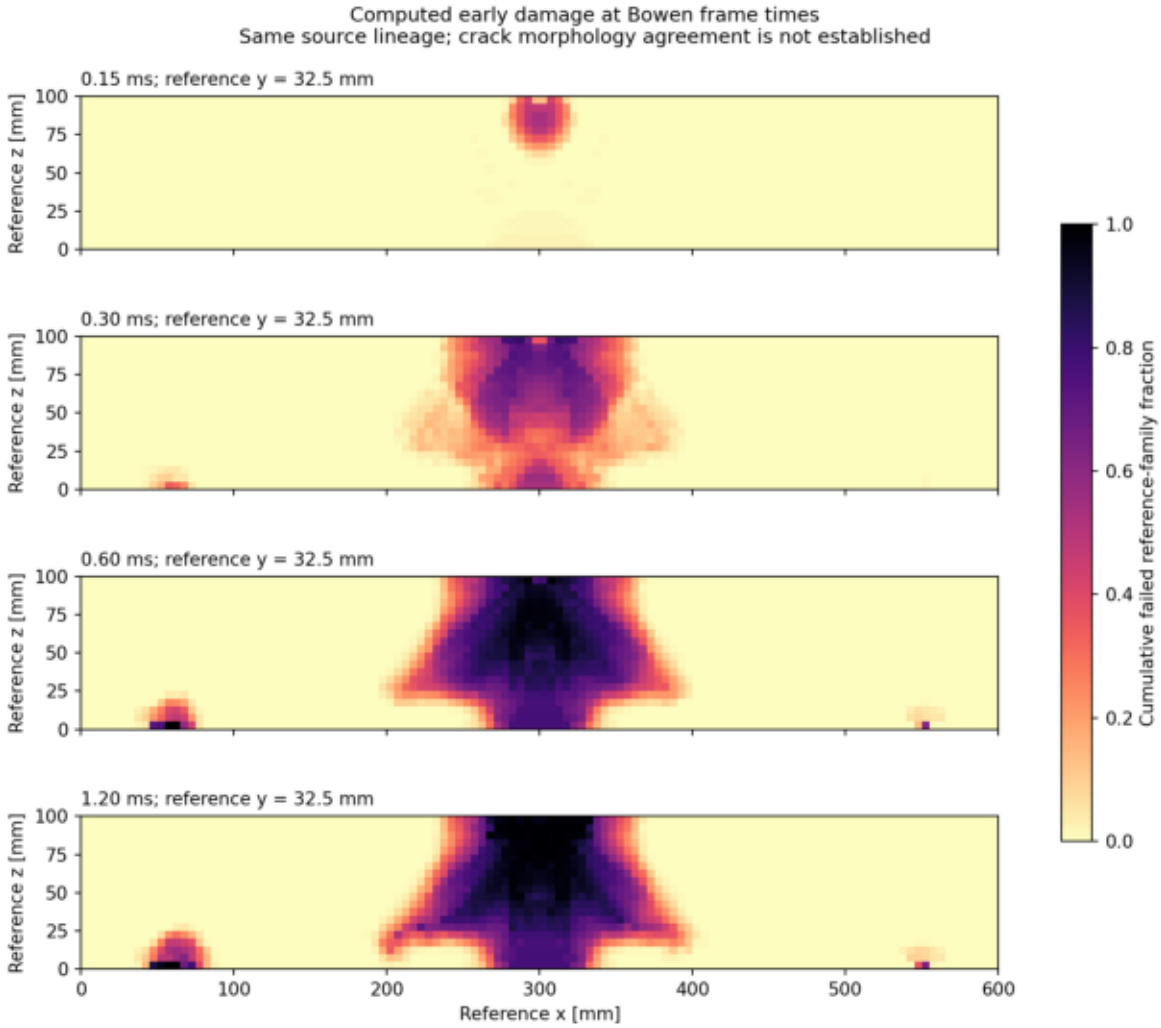
Coarse energy is sampled every 100 steps. Its last scheduled sample is step 66,600 / 19.98 ms; the final step 66,667 is intentionally unsampled, represented by NaN. All 666 scheduled samples and all final numerical state fields are finite. The final summary’s NaN energy field is not evidence of a nonfinite mechanical state.

Coarse energy at 19.98 ms	Value [J]
kinetic J	1801.663170
plastic dissipation J	157.157608
rate potential exchange J	14.491620
rupture deleted energy J	246.646961
stored energy J	6.655395
initial energy J	2239.488000

Physical drift at that sample is -41.856485 J; the separate projection-kinetic ledger is 2216.615477 J. Projection also changes potential, absent from that ledger. Small total drift or conserved momentum does not identify correct interface calibration, fracture localization or sensor response.

Independent evaluation of the unchanged final checkpoint gives stored energy 7.091248 J and physical drift -41.853385 J (-1.868882%). Concrete, PDROD, steel-rod and repulsion contributions are retained individually in the late audit.

Crack pattern: reference evidence and acceptance



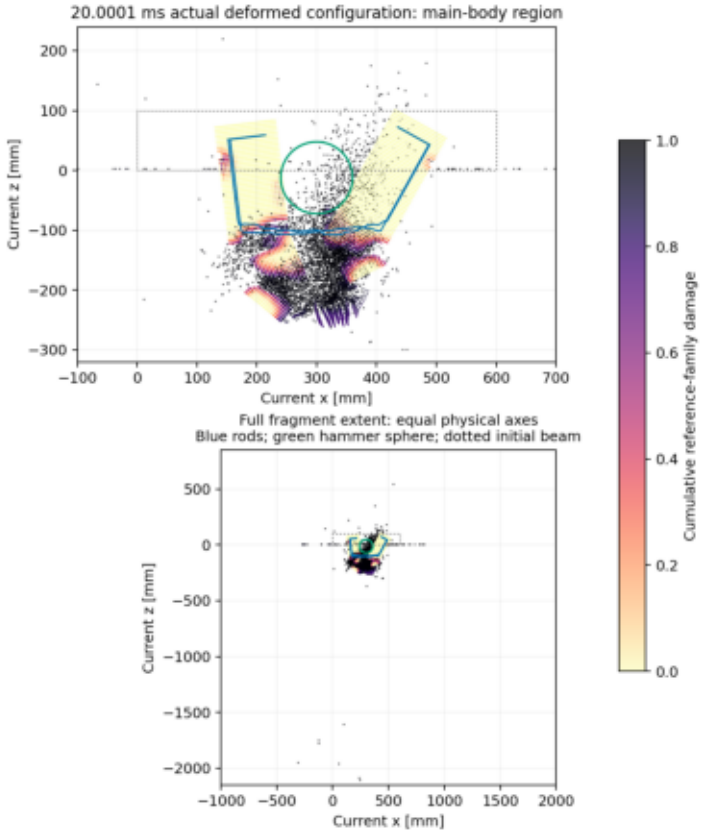
Production lineage at the exact thesis frame times: 0.15, 0.30, 0.60 and 1.20 ms. The fixed central slice uses undeformed reference coordinates, cumulative failed-family volume and a 0–1 scale. The central region becomes highly damaged. Bowen's exact slice/display mapping is unknown; the comparison is qualitative.

Bowen's H=6 m frames show two inclined shear boundaries surrounding a relatively less-damaged plug by 0.30 ms, with crushing above and local bottom spalling. The retained reconstruction at 0.75 ms shows a diffuse, heavily damaged central block; the required plug morphology has not been reproduced.

The thesis damage sequence is at 0.15, 0.30, 0.60 and 1.20 ms. "Damage evolution ended" around 1.20 ms does not mean displacement stopped. Reference-family damage, present mechanical activity, crack opening, connectivity and fragment position must remain separate diagnostics.

Bowen's 0.1701 scalar has an unmatched run/time definition. Production damage is 0.174908 at 1.20 ms; half-dt final 3 ms damage is 0.177927. Nearby global values are not morphology agreement. The exact retained early damage arrays and input-state hashes accompany the figure.

Final production damage and failed load path



Coarse checkpoint 66,667 / 20.0001 ms: 2,656 mechanical components, 2,110 isolated concrete points. Mean cumulative family damage is 0.247886; current inactive-family fraction is 0.244636.

Actual coordinates: blue rods, green hammer and dotted initial beam. The main region is enlarged above; full fragment extent is below with equal physical axes. Reference-coordinate damage slices are retained separately.

PDROD at the two midspan steel points has only 0 and 1 active samples at 3 ms, then zero at both by 6 ms. At 20 ms 96 of 270 steel nodes have no active PDROD samples; 61 of 268 rods have plastic strain, with maximum 0.30379. The steel model has no rupture state. Earlier snapshot graphs are support-connectivity upper bounds because dynamic thresholds were not saved; final checkpoint connectivity is exact.

The displacement target is exceeded without a maximum, support connectivity is lost and raw force is timestep sensitive. These establish unsuccessful reproduction, but do not uniquely identify which unresolved interface/support/rate input caused collapse.

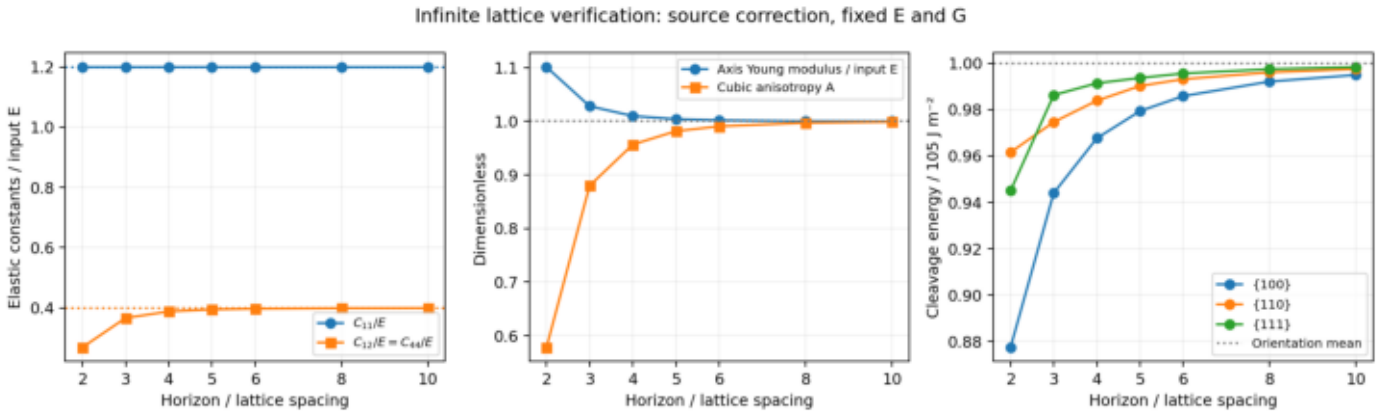
PDROD residuals are below 5×10^{-12} N and 3×10^{-12} N·m. No remaining mass/volume arithmetic error was found. Missing interface calibration and bearing/rate definitions remain essential; strengthening coupling solely to match the target would be unsupported fitting.

Cross-phase geometry: 31 concrete centres lie inside finite 3 mm rod capsules at 3 ms and 61 at 20 ms, versus zero initially. Separate steel–concrete closure contact is absent. The 3,205 restored compressive virtual samples lie outside the physical bar radius; their summed forces are not a bar-contact load. Assigning a 2.5 mm particle collision radius already creates 1,088 initial overlaps, so spacing alone cannot define a valid closure law.

Independent bulk lattice consistency audit

At the production δ/Δx = 3, the lattice is directionally biased despite nominal E = 25 GPa. The energy curvature and independent tensor agree to a maximum relative discrepancy of approximately 1.6×10^{-10}. No impact material input was changed.

δ/Δx	Axis E [GPa]	Axis ν	Cubic A
2	27.52929	0.18318	0.57820
3	25.70301	0.23403	0.87990
4	25.24733	0.24446	0.95661
5	25.10035	0.24768	0.98162
6	25.04801	0.24881	0.99056
8	25.00948	0.24964	0.99715
10	24.99952	0.24986	0.99886



Central bond forces imply C12 = C44. Production C11 = 29.99204 GPa and C12 = C44 = 9.16348 GPa give axis E = 25.70301 GPa, ν = 0.23403 and A = 0.87990. This coarse cubic anisotropy can bias directional crack propagation.

The independent orientation-mean fracture normalization is 105 J/m². At δ/Δx = 3, cleavage energies are 99.1196, 102.3371 and 103.5584 J/m² for {100}, {110}, {111}. The finite source elastic calibration at strain .001 adds a −0.02855% continuum modulus bias.

These are bulk constitutive verification results, not complete impact mesh convergence or justification to alter input E to fit the targets.

Recovered baseline and supplied Fortran evidence

The original reconstructed model completed a retained, independently checked 10,000-step / 3 ms continuation. Its full peak is 220.662764 kN; fixed bottom motion is 38.089413 mm and still downward. None of its 28 bottom probes remains bearing-connected; both steel midspan probes remain connected.

The original runner has no modern completion heartbeat schema. Completion here is established by the retained uninterrupted full history, source-checked aligned final checkpoint and recorded successful calculation, not by absence of a process. The archived baseline is an independent reconstruction, not Stage 19 and not Bowen's original executable.

Baseline checkpoint quantity	At 3 ms
Detached concrete nodes	5,348
Bottom mean / downward velocity	38.089413 mm / 14.703379 m/s
Steel midspan probes	24.076709, 23.046628 mm
Cumulative reference-family damage	0.315812

Console2.f90 is a different case: roughly 3.46 m beam, prescribed two-cylinder motion over 0.3 s, with strong damping. It is not the free 38.4 kg hammer Chapter 3 case.

Its literal 100,000-step run numerically completed with 1059.806437 kN reaction and 1300.168100 m maximum steel displacement. The yielded steel branch copies the axial accumulator into transverse components, producing a verified nonphysical force imbalance.

Five archived corrections address steel component indices, interface return-density control-volume ratio, support velocity reset, steel contact mass and cylinder tangential motion. Seven compiled native operator checks passed; the corrected 200-step elastic run completed. A full corrected 0.3 s native production result is not claimed.

The corrected different-case Fortran still cannot recover missing Chapter 3 material/interface/boundary files. Its original invalid peak must not be used as experimental agreement.

Verification, restart and remaining scientific work

The modular solver is independent of Stage 13–19 notebooks. Tests cover physical forces and masses, interpolation virtual work, force/torque conservation, rigid motion, rod yielding, contact impulse, force–energy derivatives, rupture deletion and rate-potential exchange.

The recorded full setup execution passed 62 tests in 38.43 s. Independent diagnostic and lattice suites separately passed six and eight tests. These counts describe actual executions and are not added to imply another aggregate run. The contact startup BLAS guard checked 4,096 vectors with zero mismatches.

Atomic checkpoints preserve concrete/steel motion, previous positions, flags and threshold history, rod plastic strain, hammer state and diagnostic accumulators. Restart rejects changed physics/source hashes. Complete CSV prefixes are copied only through the committed checkpoint; absolute target steps prevent silent extension errors.

The accelerated v3 package preserves physical inputs. Coarse and fine checkpoint migrations each replay 100 complete steps with bitwise equality of every saved numeric array and every history row. Separate signal/durability tests verify finishing a committed step after SIGINT/SIGTERM, discarding a partially mutated exception state and preserving a checkpoint-aligned history after forced termination.

Independent validators preserve their exact input CSV/checkpoint bytes, reject incompatible manifests and conflicting restart rows, and separate physical energy drift from the projection kinetic ledger. The current production and refinement windows have all completed; none establishes the original author’s full model.

Next scientific requirements

Resolve source-defined Chapter 3 interface and supports where evidence exists; test complete contact geometry and a justified observation operator; establish full impact mesh/time sensitivity; compare time-matched shear-plug morphology and a fixed experimental displacement sensor. Original solver/input and measurement definitions remain missing. No arbitrary fitting is authorized by these gaps.

Continuing ballistic fragment motion solely to reach a desired number is not useful validation. The completed 20 ms calculation is retained as a failed candidate with diagnostics, rather than relabelled as Bowen’s result.

The large Peridynamic articles.zip remains uninspected because the available download tool rejects its size. This report does not claim exhaustive review of every supplied archive.

Source-pinned report provenance

This PDF uses aligned final retained histories/checkpoints, exact package hashes and independent validator metrics. Its companion JSON and snapshot directory retain all five run inputs, manifests, CSV prefixes, native checkpoints, figures, completion evidence and scientific audit artifacts.

Production 20 ms: pdbeam_v3; 66,667 steps; checkpoint retained.

Checkpoint SHA256: a1cec9aa7c078260509054fe630ffec4dedaed47221fc5b9d267f4d33c5f2f60

History SHA256: 513d8bb6a3959e703bb7e3f0c8be0a29ead02db0e8d2899a7016d3f22b31ac8d

Half-dt 3 ms: pdbeam_v3; 20,000 steps; checkpoint retained.

Checkpoint SHA256: 2ba567ca45224dfcaf967971c663019d09b598aee3fd3c81c662868bb617c6a5

History SHA256: 9097bbddda147267c7e60e07aa5039a81507142daf2efdd7e94b14fa048ba411

Quarter-dt .3 ms: pdbeam_v3; 4,000 steps; checkpoint retained.

Checkpoint SHA256: ffe42ca84a1c87b758c398fb2854f1dfb4da1db1c0087e6546c3eafed6ad056b

History SHA256: d0d159c1492ece42eb728c9a6a9c23cfb4d6a6869fad91319a1d7dd8379767a3

Eighth-dt .3 ms: pdbeam_v3; 8,000 steps; checkpoint retained.

Checkpoint SHA256: 4a2d15b23697a4bacf435a79402e94cea66ee4721480ab4911fafa7135ef7078

History SHA256: f0f7b6ae87e1d44f1cb47343a82aaa3658489643a64eb8125751f85545204be4

Original 3 ms: pdbeam; 10,000 steps; checkpoint retained.

Checkpoint SHA256: b5f2499edcfc079a06d90d758423474457ad07e5ab95b23dbed24c4fcf778f52

History part 1/2 SHA256: e466fe10312315965f94b1f560a5292235f5a380e66387b41f627874e15ac74c

The original history also requires its archived 1–5,000 prefix. All prefix hashes and complete records are preserved in JSON metadata. A source/configuration hash identifies the implemented scientific operator; it does not establish equivalence to the missing author code.

Run completion observations at report time

corrected_20ms_accelerated: completed requested window.

corrected_halfdt_3ms: completed requested window.

corrected_quarterdt_0p3ms: completed requested window.

corrected_eighthdt_0p3ms: completed requested window.

recovered_original_3ms: completed 3 ms window (retained history/checkpoint evidence).

Rebuild from /workspace/bowen_continued:
OPENBLAS_NUM_THREADS=1 .venv/bin/python research/reporting/build_report.py

This generator refuses changed source/evidence bytes, writes the PDF atomically, and does not alter solver code, raw results or the GitHub checkout. The public PDF publication is a separate reviewed action.